Beyond Explicit Ratings: Reaction Time as an Implicit Measure of Category Structure

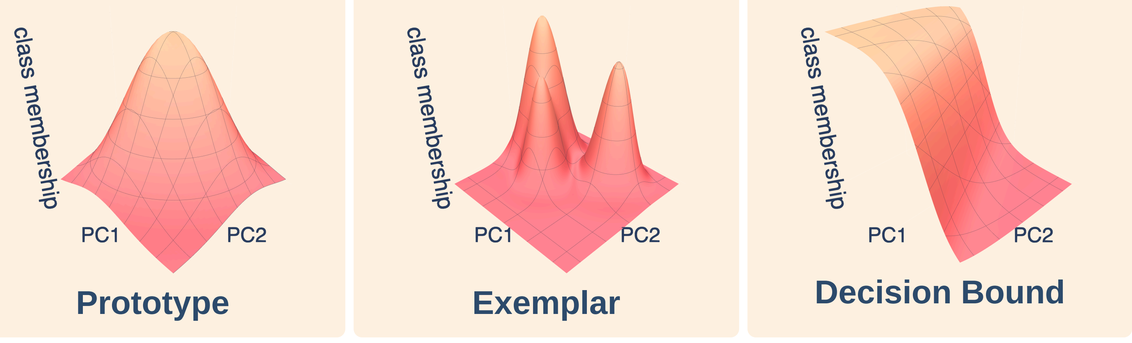
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Introduction

Humans classify complex visual stimuli quickly and effortlessly. Yet the computational mechanisms underlying this ability remain unresolved. Research on these mechanisms centers on two central questions:

1. What is the *representational space* in which categories are defined?
2. What is the *internal structure* of categories within that space?

THREE CATEGORY THEORIES



MOTIVATION & HYPOTHESIS

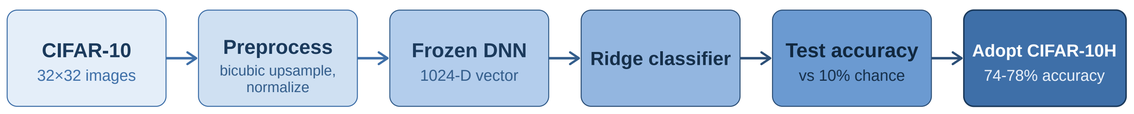
Explicit typicality ratings risk demand bias. We instead use reaction time (RT) as an implicit proxy: less typical images take longer to classify. We correlated DNN-derived typicality from 30 model combinations with human RTs, and validated RT as a reliable measure via Hick's Law.



Methods

DATASET - CIFAR-10H

Selected for its rich behavioral data - including reaction times and 51 human classification labels per image. Its low resolution (32×32 px) required verifying that informative representations could be extracted:



All models scored far above chance, confirming informative representations.

NEURAL NETWORK MODELS

Vision-Language:

CLIP ViT-L | SLIP ViT-L (CLIP) | SLIP ViT-L (SLIP) | SigLIP2

ImageNet Classifiers:

Google ViT-L | ConvNeXt-L | Inception-V3 | CORnet-RT

Self-Supervised:

DINOv2 ViT-L | SLIP ViT-L (SimCLR)

HYMAN-HICK'S LAW

Hyman-Hick's Law predicts longer RTs under greater stimulus uncertainty. We quantify per-image uncertainty as the entropy of the human label distribution, and test whether higher entropy is associated with longer mean RTs.

Hyman-Hick's Law

$$RT = a + b \cdot H$$

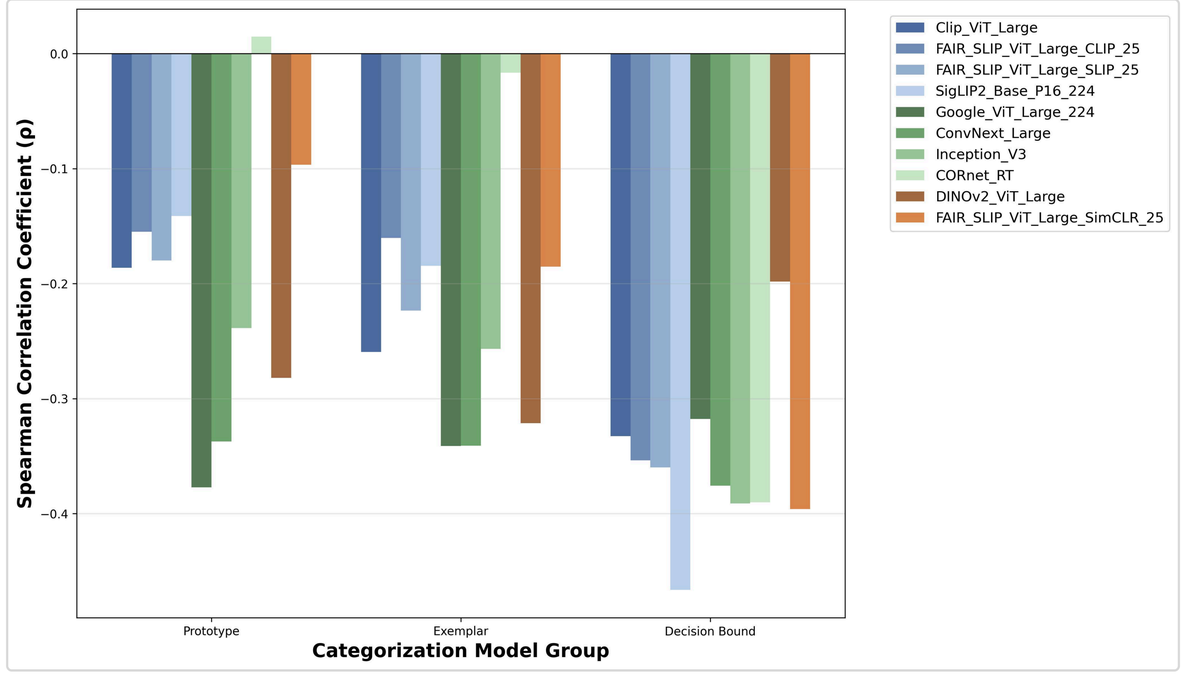
Shannon Entropy (Miller-Madow corrected)

$$H(X) = -\sum p_i \ln(p_i) + \frac{(K\text{-}1)}{2N}$$

Results

SPEARMAN CORRELATIONS

Across nearly all 30 combinations, typicality correlated negatively with RT, strongest for Decision-Bound + SigLIP2.

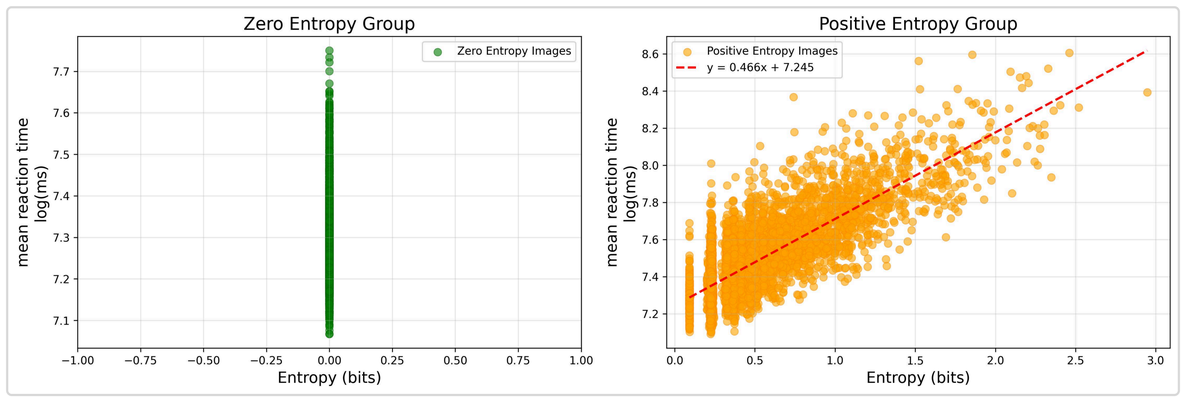


HYMAN-HICK'S LAW VALIDATION

RTs are plotted as log(mean RT) to normalize their skewed distribution and stabilize the mean. As Hyman-Hick's Law predicts, RTs rise with label entropy, confirming RT is a reliable measure.

0.689
Spearman ρ
high-entropy subset

p<.0001
significance



Conclusions

- RT correlates negatively with model typicality, as predicted — typical images are classified faster, so model typicality captures this implicit measure.
- RT is a valid implicit signal: Hyman-Hick's Law holds on log-scaled RTs, confirming the data and pipeline are sound.
- Model families differ in how well they capture category structure: classification models are less responsive to the category structure theory, whereas vision-language and self-supervised models show a significantly greater response.
- Future work: test whether any single combination significantly outperforms the rest; examine the link between explicit (ratings) and implicit (RT) measures, and whether dataset choice drives the difference.

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